

When to put call takers on the phones

A 911 call-taker scheduling proposal built on Anton's call-volume forecast · Oakland Police Department · PITC Fall 2026 · Draft for team review · Revision 2 (Sep 30), see last page

The answer. Anton's forecast shows 911 calls follow the same curve every week: quiet at 4am, nearly double by 9am, flat and busy until 9pm. OPD's schedule doesn't follow that curve. Moving shift starts to **4am, 8am, 12pm, 2pm and 6pm** and running **13 ten-hour call-taker shifts a day instead of ~10.5** (about **+4.4 full-time call takers; +4 to +6 depending on assumptions**) gets OPD from **77% to about 91%** of 911 calls answered within 15 seconds on an average day (the state standard is 90%). The weakest hour still reaches only 62%, so Option C below is the route to 90% in nearly every hour.

Units. A "call taker" here is a 911-answering position counted the way the City Auditor counts it (Exhibit 19: total minimum staffing minus the 4 radio and service-desk positions, divided by 2). Radio and service-desk positions are not modeled. Adding 4.4 call-taker FTE could therefore mean up to about twice as many dispatcher hires if OPD keeps staffing its non-emergency lines at the same ratio, so whether this fits within OPD's 12 open dispatcher positions is **not yet established** and needs OPD's roster.

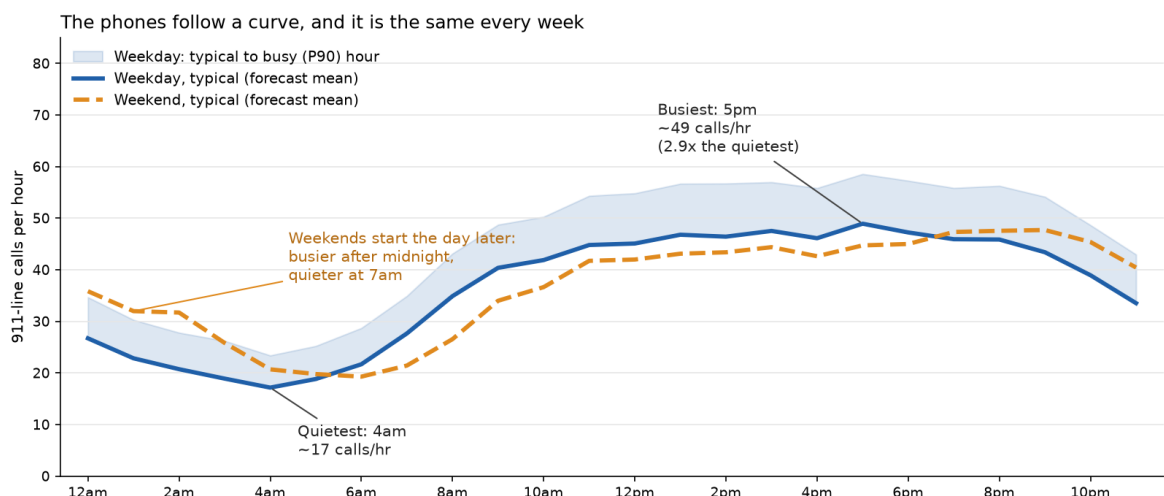
1. The problem: callers are waiting

California's standard is that **90% of 911 calls are answered within 15 seconds**. OPD answered **54%** in 2024 and **73 to 79%** in Jan to Jul 2026, below the standard in every month. Every call past 15 seconds is a caller listening to ringing in an emergency. The question for this proposal: **how many call takers, starting when, closes that gap?**

2. What Anton's forecast tells us: the day has a shape

Anton's model forecasts public calls for every hour, issued 48 to 71 hours ahead, which is the window in which shifts can still be changed. We use its back-tested forecasts for Jul 2025 to Jun 2026 (what it would have said each day) and convert them to 911-line calls (see appendix). Four things in it drive the schedule:

- **The curve is steep.** Calls bottom out at about 4am (~17 an hour) and peak around 5pm (~49 an hour), **2.9 times** as many.
- **The morning climb is fast.** Weekday volume nearly doubles between 6am and 9am, so people must already be at their desks before it starts.
- **Weekends run on a different clock.** About 9 more calls an hour at 1am and 6 fewer at 7am than on weekdays, although total daily volume is the same (~880 calls).
- **The forecast is trustworthy.** Across the 8,756 hours tested (Jul 2025 to Jun 2026), his P90 (the level 9 of 10 hours stay under) held **91.5%** of the time. We plan on the P90, because an unanswered call costs far more than an idle seat.



Anton's hourly forecast, Jul 2025 to Jun 2026, issued 48 to 71 hours ahead; converted to 911-line calls. His P90 held in 91.5% of the 8,756 hours tested.

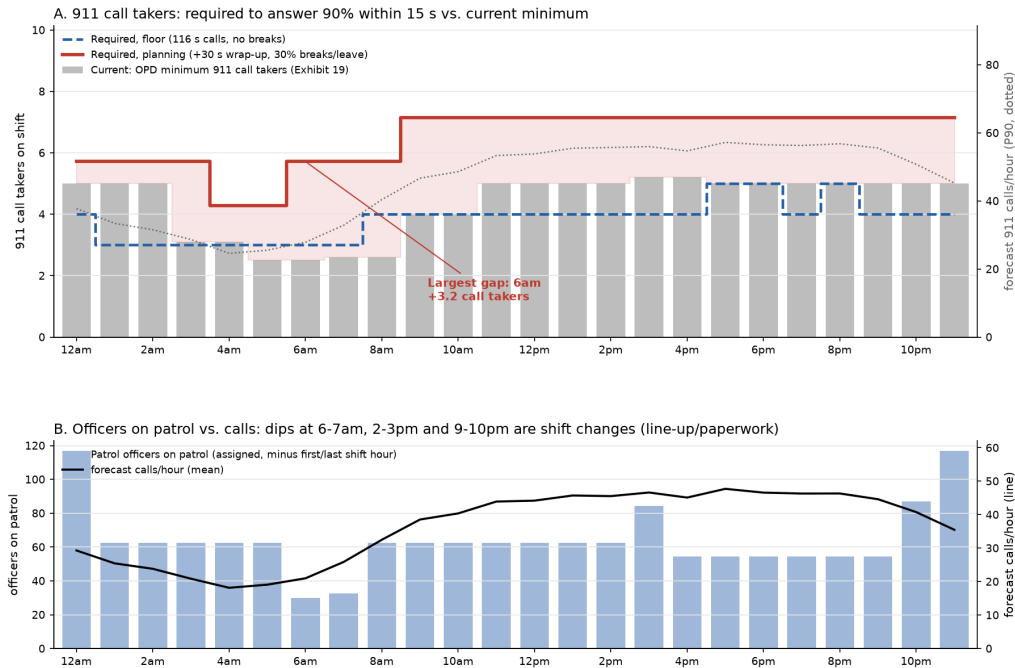
Read it: the blue line is a typical weekday, the shaded band runs up to a busy (P90) hour, and the orange line is a typical weekend.

3. Today's schedule doesn't follow the curve

OPD's minimum is about 5 call takers for most of the day, and **2.5 to 3 between 4 and 9am**, just as calls double. With realistic breaks and after-call work, the standard needs about **5.7 call takers overnight and mornings and 7.1 from 8am on**. The biggest gap is at **6am, 3.2 call takers short**. That is also the hour officers on patrol drop to about 30 (panel B), so the thinnest hour is thin at both ends of the radio. Without breaks and wrap-up (blue dashed line) the minimum would be enough except 5 to 8am, but OPD actually answered only 73 to 79% of calls in time, so that floor understates the need.

With realistic breaks and wrap-up, OPD's minimum is 2-3 call takers short at most hours — worst 6-10am

Average day, Jul 2025 – Jun 2026 back-tested forecast (Anton's model, P90). Erlang C, CalOES standard: 90% of 911 calls within 15 s. Without breaks/wrap-up (blue) only 5-8am is short.



Sources: forecast cache/bt_glm_a36ba550a4.pk; 911-line conversion from 2024_answer_time.csv; talk time = San Francisco 2023 (audit Exhibit 49); current = audit Exhibit 19; officers = opd_patrol_officers_by_shift_day.csv. Call takers = the audit's estimate of 911-allocated call takers (total minimum staffing - 4 radio/service-desk / 2); radio and service desks are not modeled.

Read it: grey bars are today's minimum, the red line is what the standard needs, and the pink area is the gap. Panel B leaves out each patrol shift's first and last hour (line-up and paperwork).

4. Why shift timing is the lever

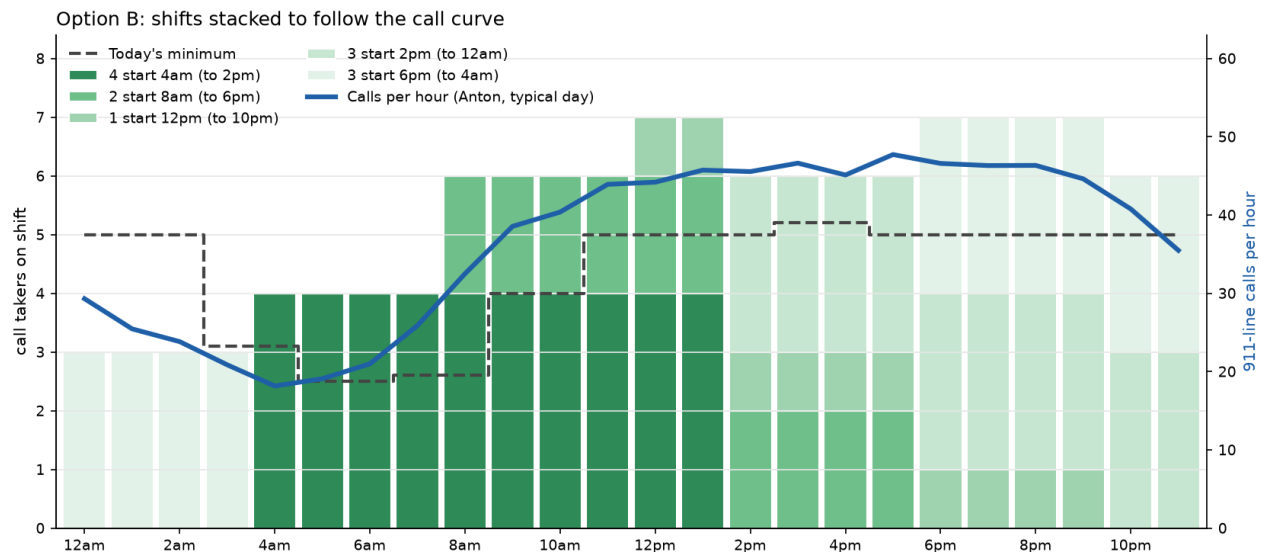
Call takers work **4 ten-hour shifts a week**, so one hire covers a block of the curve, not a single hour. That makes **shift start time** as important as headcount. Re-timing today's shifts alone (Option A) lifts the answer rate only from 77% to 81%. Reaching the standard takes both: more shifts, placed where Anton's curve is steepest.

Option	Call-taker shifts/day	Added full-time call takers	Est. answered ≤15 s	Worst hour	Shift starts (people per day)
Today (Exhibit 19)	~10.5	—	77% (actual 2026: 73–79%)	—	Current minimum schedule
A. Re-time + ~1 hire	11	+0.9	81%	65%	2am ×2, 6am ×3, 12pm ×2, 4pm ×3, 8pm ×1
B. Meet the standard	13	+4.4	91%	62%	4am ×4, 8am ×2, 12pm ×1, 2pm ×3, 6pm ×3
C. 90%+ nearly every hour	16	+9.7	97%	93%	4am ×3, 8am ×4, 2pm ×4, 6pm ×3, 10pm ×2

Every option keeps at least 3 call takers on the phones every hour and uses at most 5 shift start times. Full-time equivalents (FTE) = added call-taker hours per day × 7 ÷ 40. **Sensitivity:** Option B needs 13 shifts (+4.4 FTE) with 15 to 60 s of after-call work, or if today's true answer rate is 79%; it needs 14 shifts (+6.2 FTE) with no after-call work, or if today's true rate is 73%. Option B averages 91% over the day but its weakest hour reaches 62%.

Option B, shift by shift. Follow the curve from left to right:

- **4am start (4 people)** are at full strength before the 6 to 9am climb, when today's schedule is at its thinnest.
- **8am and 12pm starts (3 people)** add cover for the morning rise and the midday plateau.
- **2pm start (3 people)** takes over as the 4am group leaves, through the evening peak.
- **6pm start (3 people)** carries the evening and the quiet night to 4am, then hands over to the 4am group.



Read it: each green layer is one shift start. The blue line is calls per hour; the dashed line is today's minimum.

5. Where Anton's forecast keeps the roster honest

A fixed roster is built on a typical day, but days differ. Looking at actual calls over the same year, the same shift can see very different loads:

Shift (10 h)	People	Typical calls in the shift	Quiet day to busy day (10th to 90th percentile)	Busy-day swing vs typical
4am to 2pm	4	~327	291 to 364	+11%
8am to 6pm	2	~427	375 to 473	+11%
12pm to 10pm	1	~457	412 to 498	+9%
2pm to 12am	3	~440	403 to 493	+12%
6pm to 4am	3	~351	306 to 436	+24%

The **evening-into-night shift swings most**. It is a weekend effect: a 6pm shift sees about **415 calls on Fridays and Saturdays versus about 335 Monday to Thursday** (23% more), and 35 of its 37 busiest days are Fridays or Saturdays. Because Anton's forecast for any day is available **48 to 71 hours ahead**, there is still time to act. We propose three layers:

Layer 1. A base roster (Option B) built on the typical curve.

Layer 2. A Friday/Saturday variant with extra evening cover and later starts, since those nights run ~23% hotter and weekends start the day slowly (not yet modeled; next step).

Layer 3. A two-day-ahead check. Each morning, compare Anton's P90 for the day after tomorrow with the roster. Where the P90 shift load is above what the roster covers, fill the gap with an overtime or swap request while it can still be arranged.

6. What we're asking for

Adopt Option B: 13 call-taker shifts a day, starting at 4am, 8am, 12pm, 2pm and 6pm, about **+4.4 full-time call takers**. Estimated result: **91% of 911 calls answered within 15 seconds** on an average day, up from 77%. Planning range: +4 to +6 FTE depending on assumptions. Option C (+9.7 FTE, 97%) is the route to 90% in nearly every hour. Whether either fits within the 12 open dispatcher positions depends on how call-taker FTE convert to dispatcher hires (see the units note on page 1). To firm this up we need four things from OPD: after-call work time per call, real breaks and leave, the actual dispatch-center shift sheets and roster, and how dispatchers split between the 911 and non-emergency lines.

Appendix: how we got there

Step	Input	Source
1. Calls per hour	Anton's back-tested forecast (all public CAD calls: 911, non-emergency phone, text), planning on the P90. Converted to 911-line calls with the hourly ratio seen in Jan to Jun 2024 (overall 0.99x, 0.90 to 1.09 by hour). His own next step is to forecast phone-system call arrivals directly.	oakland_911_forecast.ipynb, final model back-test (Box: cache/bt_glm_a36ba550a4.pkl); 2024_answer_time.csv
2. Time per call	116 s talk , the San Francisco benchmark. Oakland's own 2024 figure is 112 s, so the proxy is close.	City Auditor 911 audit (Oct 2025), Exhibit 49
3. Erlang C	Smallest number of call takers so that 90% of callers wait 15 s or less at that hour's volume.	Standard call-center queueing formula
4. Current staffing	OPD's minimum 911 call takers by hour: 2.5 to 5.2. This is the auditor's estimate: total minimum staffing minus 4 radio/service-desk positions, divided by 2.	Audit Exhibit 19 and its note (2023 schedule, read off the chart)
5. Shift plan	Integer program: best 10-hour shift starts (OPD works 4x10s) for each staffing level.	dispatcher_scheduling_proposal.py

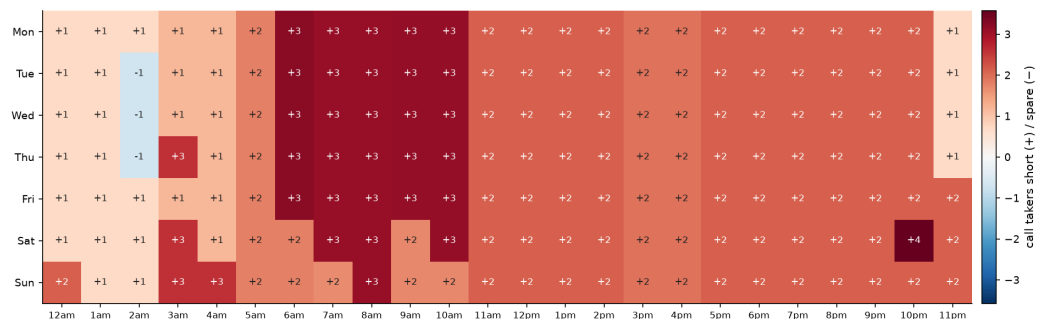
Two assumptions Anton asked about

Work after the call (can't pick up right away)	Added as +30 s per call of after-call work (CAD entry), so handling time is 146 s instead of 116 s. This is a placeholder; OPD's CAD entry time would replace it.
Breaks, bathroom, coffee, talking to colleagues	Modeled as 30% shrinkage : only 70% of scheduled people are on the phones at any moment (also covers training and leave; typical contact-center range is 25 to 35%). Calibration, not validation : the two assumptions were chosen so today's schedule comes out at 77%, in line with OPD's actual 73 to 79% (Jan to Jul 2026). Other pairs that also give 77% lead to +4.4 to +6.2 FTE (section 4). Without them the model says the minimum is enough except 5 to 8am, which the real data shows isn't true.

Where the gaps are, by weekday and hour

Where the gaps are: red = understaffed, blue = more than needed

Required (planning scenario) minus current minimum, by weekday and hour. Forecast P90, Jul 2025 – Jun 2026.

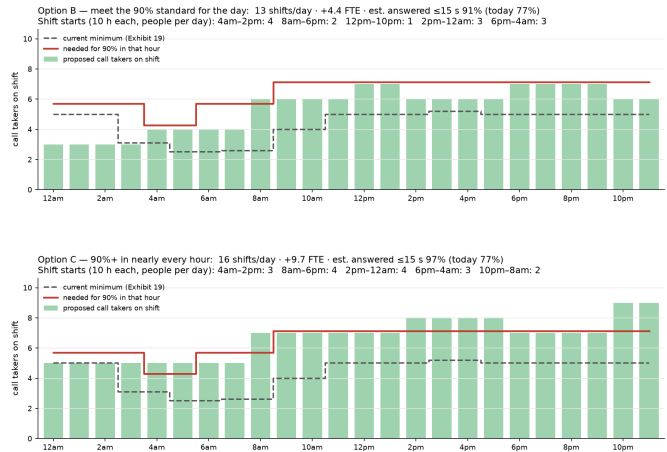


Current minimum (Exhibit 19) is the same every day; required varies by day from the forecast. Planning = 116 s talk + 30 s wrap-up, 30% shrinkage (calibrated so today's schedule gives 77%). Call takers = audit's 911-allocated estimate (Exhibit 19).

Read it: red = understaffed, blue = more than needed. The gap is widest on weekday mornings, 6 to 10am (+3), and Saturday at 10pm (+4).

Options B and C compared

Shift proposal: add daytime and evening shifts starting 4am, 8am, 2pm and 6pm
Optimised 10-hour shift starts (±5 start times, never fewer than 3 call takers) for the average day, planning scenario.



FTE = added call taker hours/day × 7 ÷ 40. Answer rates are Erlang C estimates on the forecast mean with 30% shrinkage and 146 s handling time; the assumptions are calibrated so today's schedule gives 77%, in line with the 73-79% measured Jan-Jul 2026 (not independently validated). Call takers = audC's 911-allocated estimate.

Limits and next steps

Limit	What would fix it
After-call work (30 s) and shrinkage (30%) are assumptions, calibrated so today's schedule gives 77%.	OPD data request: CAD entry time per call, and actual breaks and leave.
'Call takers' are the auditor's estimate of 911-allocated positions; hires may be up to about twice as many dispatchers.	OPD roster and how dispatchers split between the 911 and non-emergency lines.
Anton's forecast counts all public CAD calls; we convert to 911-line calls with a Jan to Jun 2024 ratio.	Phone-system (call-arrival) logs, as Anton's next steps suggest.
We use hourly P90s, not Anton's shift-peak forecasts (NB GLM + day shock, 9.1% missed peaks at P90).	Once shifts are final, put them in his notebook (his SHIFTS setting) and use its shift-peak quantiles.
Current staffing is the auditor's 2023 minimum, not who actually works each hour.	OPD data request: dispatch-center shift sheets and roster.
Covers the 911 line only; call takers also answer 10-digit lines.	Add the 10-digit line volume and handle time from OPD's phone system.
Erlang C assumes callers never hang up and calls arrive at random; real waits are burstier.	Re-check against hourly 2025 to 2026 phone data when it arrives.
The plan is for an average day from back-tested forecasts; a weekend variant and the two-day-ahead trigger are not yet modeled.	Model weekday and weekend rosters; test how often the P90 check would have fired in the forecast year.

Revision notes: what changed since the first version shared

- **Units explained.** "Call takers" are the auditor's 911-allocated estimate. The claim that the plan "fits inside the 12 open dispatcher positions" is removed until OPD's roster shows how call-taker FTE convert to hires.
- **Range added.** +4.4 FTE is the base case; +4 to +6 depending on the after-call work and answer-rate assumptions. The two added assumptions are described as calibrated, not validated.
- **Numbers corrected.** Weekend vs weekday: +9 calls/hour at 1am and -6 at 7am (was 10 and 7). Friday/Saturday 6pm shift: about 415 vs 335 calls, 23% more (was 400 vs 325, about 20%).
- **Clearer results.** The 91% is a daily average (weakest hour 62%). Without breaks and wrap-up only 5 to 8am is short, and why that floor understates the need.
- **Anton alignment noted.** Forecast is a back-test; the CAD-to-911 conversion and the unused shift-peak forecasts are listed as limits.